Erdi Sayar

Reinforcement Learning & Autonomous Drone Systems Researcher

erdi.sayar@tum.de | LinkedIn | GitHub | Portfolio | Google Scholar |
Nationality: Turkey & Germany

Education

Technical University of Munich — Munich, Germany
Ph.D. Computer Science — *2020 – 2025*
Thesis: Optimization control and sample efficiency in reinforcement learning for robotic tasks.
First Examiner: Prof. Dr. Alois Knoll (TUM) Second Examiner: Asst. Prof. Andreea Bobu (MIT)

RWTH Aachen University — Aachen, Germany
M.Sc. Electrical Engineering, Information Technology and Computer Engineering; ***GPA: 1.5/1.00*** — *2018 – 2020*
Thesis: Anomaly detection in unknown environments using sensory agents
Advisor: Prof. Dr. Gerd Ascheid Grade: 1.0

Bochum University of Applied Sciences — Bochum, Germany
B.Sc. Mechatronics Engineering; ***GPA: 1.36/1.00*** — *2013 – 2014*

Kocaeli University — Kocaeli, Türkiye
B.Sc. Mechatronics Engineering; ***GPA: 3.82/4.00*** — *2010 – 2015*
Valedictorian, ranked first in the faculty of engineering. Graduated with top honors.

Experience

Postdoctoral Researcher
Paderborn University — *09/2025 – present* — *Paderborn, German*
- *Research in reinforcement learning, vision-language-action (VLA) models, and safe Generative AI.*

Teaching
- *Advanced System Theory*
- *Nonlinear control of autonomous and robotic systems*

Research Associate
Technical University of Munich — *10/2020 – 08/2025* — *Munich, Germany*
- *Research on reinforcement learning, diffusion models, and curriculum learning.*

Teaching
- *Kognitive Systeme (IN2222) – Summer 2023, 2024 (lecturer and exercise lead)*

Student Assistant
RWTH Aachen University — *10/2018 – 04/2019* — *Aachen, Germany*
- *Research on anomaly detection in unknown environments using miniaturized sensory agents.*

Research Assistant
Izmir Institute of Technology — *02/2017 – 02/2018* — *Izmir, Türkiye*

Teaching
- *ME352 System Analysis and Control*
- *ME242 Applied Mathematics for Engineers*

Research Assistant
Turkish-German University — *02/2016 – 02/2017* — *Istanbul, Türkiye*

Journal Publications

1. **E. Sayar**, X. Gao, Y. Hu, G. Chen, A. Knoll. *Toward coordinated planning and hierarchical optimization control for highly redundant mobile manipulators.* ISA Transactions, 2024.

2. **E. Sayar**, G. Iacca, A. Knoll. *Curriculum learning for robot manipulation tasks with sparse reward through environment shifts.* IEEE Open, 2024.

Conference Publications

3. **E. Sayar**, E. Das, J. W. Burdick, A. Knoll, E. Kayacan. *D-SafeMPC: Diffusion-Driven Safe Model Predictive Control with Discrete-Time Control Barrier Functions.* IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

4. V. H. Dang, K. Rajendran, **E. Sayar**, E. Kayacan. *Exp2VLA: Enabling Vision–Language–Action for Drone Navigation from Expert Demonstrations.* IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), 2026.

5. H. Sahin, V. H. Dang, **E. Sayar**, A. Yegenoglu, E. Kayacan. *Fuzzy Logic Theory-based Adaptive Reward Shaping for Robust Reinforcement Learning (FARS).* IEEE World Congress on Computational Intelligence (WCCI), 2026.

6. F. Oruç, **E. Sayar**, G. Iacca, A. Knoll, E. Kayacan. *Palette Inpainting Diffusion Curriculum Reinforcement Learning (PIDCRL).* IEEE World Congress on Computational Intelligence (WCCI), 2026.

7. **E. Sayar**, G. Iacca, O. Oguz, A. Knoll. *Diffusion-based Curriculum Reinforcement Learning.* Neural Information Processing Systems (NeurIPS), 2024.

8. **E. Sayar**, G. Iacca, A. Knoll. *Multi-Objective Evolutionary Hindsight Experience Replay.* The Genetic and Evolutionary Computation Conference (GECCO), Melbourne, Australia, July 2024.

9. **E. Sayar**, Z. Bing, C. D'Eramo, O. Oguz, A. Knoll. *Contact Energy Based Hindsight Experience Prioritization.* IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.

10. **E. Sayar**, V. Vintaykin, G. Iacca, A. Knoll. *Hindsight Experience Replay with Evolutionary Decision Trees for Curriculum Goal Generation.* European Conference on Genetic Programming (EuroGP), Wales, United Kingdom, April 2024.

11. C. Sariman, A. Hallawa, **E. Sayar**, A. Peine, L. Martin, A. Schmeink. *Exploration of unknown environments via evolution of behavioral and morphological properties of miniaturized sensory agents.* The Genetic and Evolutionary Computation Conference (GECCO), Boston, USA, July 2022.

12. **E. Sayar**. *Assistive Handwriting Haptic Mechanism Using Deep Learning Speech Recognition.* European Conference on Mechanism Science (EuCoMeS), Cluj-Napoca, Romania, 2020.

13. **E. Sayar**, M. Ertunc. *Fuzzy logic controller and PID controller design for aircraft pitch control.* European Conference on Mechanism Science (EuCoMeS), Aachen, Germany, September 2018.

Selected Projects & Open-Source Work

- **Diffusion-based Curriculum RL** – *PyTorch implementation of diffusion-based curriculum learning for sparse-reward manipulation.*
- **Mobile Manipulator Control** – *Hierarchical optimization framework for redundant mobile manipulators (ROS, MuJoCo).*
- **Hindsight Experience Prioritization** – *Contact-energy-based HER baselines on Fetch and Hand manipulation benchmarks.*

Mentoring

- **M.Sc. Students:** Md Siyam Sajeeb Khan, Xiyuan Wang, Efe Kaya, Faruk Oruc, İlker Ayçiçek, Alexander Höhn, Ramandika Pranamulia.
- **Undergraduate Students:** Vladislav Vintaykin, Khalil Chikhaoui, Emre Öztas, Mohamed Bouhali, John F. Stewardson.

Academic Service

- **Special Session Organizer.** "Generative Foundation Models for Robotics: From Language and Vision to Embodied Action," IEEE World Congress on Computational Intelligence (WCCI), 2026.
  wccigenai2026.github.io
- **Summer School Co-Organizer.** "Foundational Models Meet Robotics: From Language and Vision to Action," IEEE CIS Summer School, Paderborn University, 2025.
  upb-rat.github.io/ieee-summer-school

Skills

Languages: C/C++, Python, MATLAB & Simulink, Maple, Linux, HTML, CSS, Javascript, LaTeX

Technologies: Git, Docker, PyTorch, TensorFlow, JAX, ROS, Unity, MuJoCo, Isaac Sim, gRPC, Ansys, Solidworks, Blender

Languages: Turkish (Native), English (C1), German (B2)

Additional Activities & Achievements

- Fall School on Model Predictive Control and Reinforcement Learning, Freiburg, 2025.
- Certified A1/A3 Open Category Drone Pilot (EU Regulation).
- DLR & TU Braunschweig ASDA Spring School on Innovative Drone Applications, 2025.
- Ranked 1st in the Faculty of Engineering, Kocaeli University (over 1000 students).
- ROS Summer School Certificate at FH Aachen.
- Voluntary Work for Assisting Elderly People with Shopping in Garching/Munich.